



Existence of Kähler Algebras with Chow Polynomials as Hilbert Series

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Can we define a ring with nice algebraic properties for every poset?

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Joint work with Adam Schweitzer (KTH).

Chow rings

The matroid Chow ring

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where

$$I_M = \left\langle \sum_{F \ni i} x_F : i \in E \right\rangle \quad J_M = \langle x_F x_G : F, G \text{ incomparable} \rangle,$$

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Monomials that do not vanish are supported on chains of flats.

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where

$$\emptyset = F_0 < F_1 < \cdots < F_s \leq E \quad \text{and} \quad 1 \leq \ell_i \leq \mathrm{rk}(F_i) - \mathrm{rk}(F_{i-1}) - 1.$$

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$$H_M(t) = \mathrm{Hilb}_{\mathrm{CH}^\bullet(M)}(t) = \sum_{i \geq 0} \dim_{\mathbb{R}}(\mathrm{CH}^i) t^i = \sum_{i \geq 0} |\mathrm{FY}^i| t^i.$$

We write FY^k for the set of Feichtner–Yuzvinsky monomials of degree k .

Only rank-jumps of at least 2 give contributions.

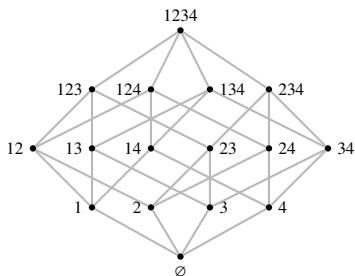
Example

$$\emptyset = F_0 \preceq F_1 \preceq \cdots \preceq F_s \leq [4],$$

$\leadsto$

$$x_{F_1}^{\ell_1} \cdots x_{F_s}^{\ell_s},$$

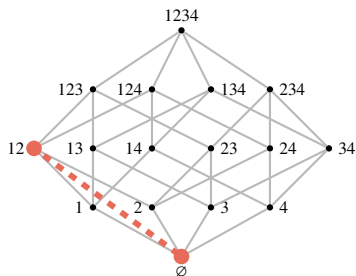
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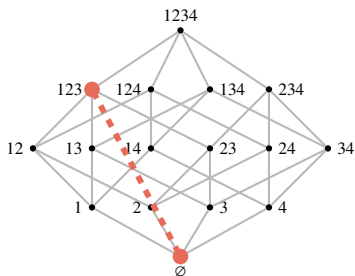


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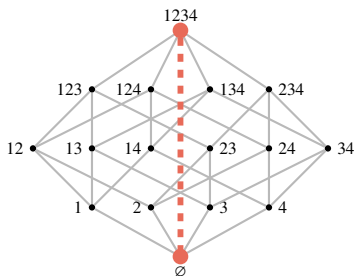


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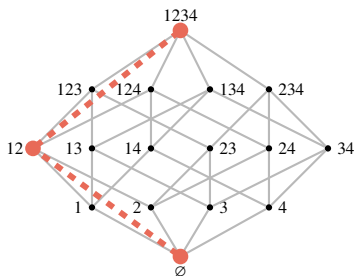
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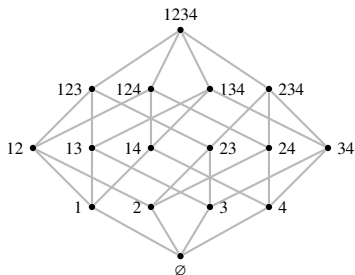
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$$H_{U_{4,4}}(t) = 1 + 11t + 11t^2 + t^3.$$

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$CH^\bullet(M)$ satisfies Poincaré duality, Hard Lefschetz and Hodge–Riemann relations.

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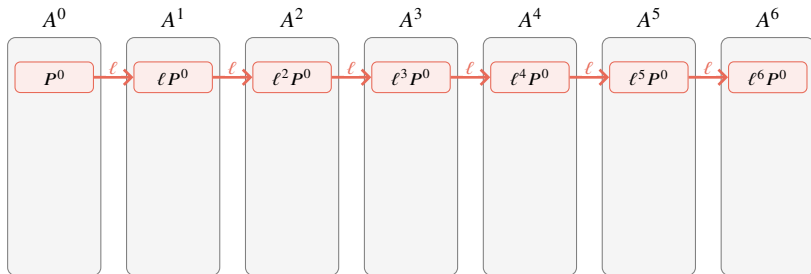
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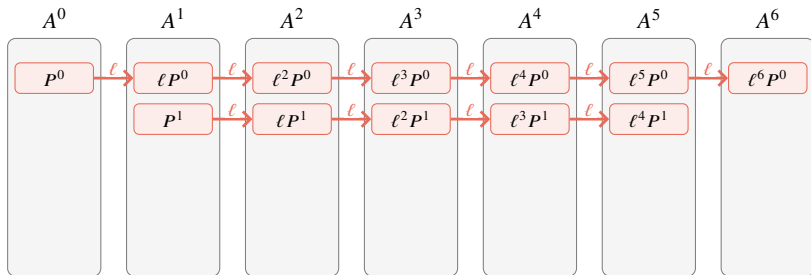
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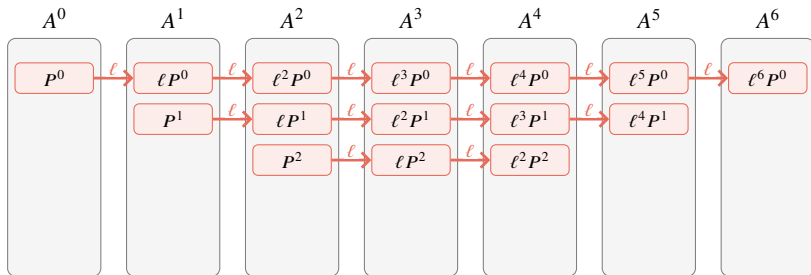


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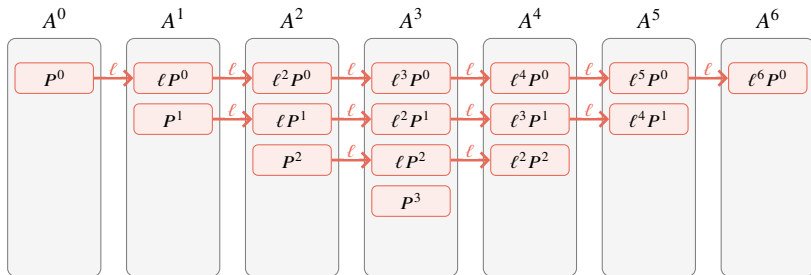


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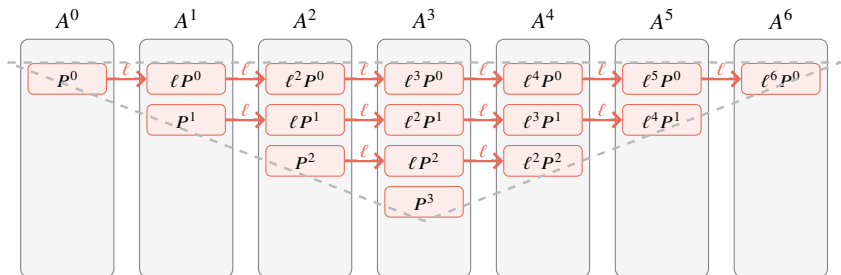


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Hilbert series and combinatorial consequences

$$H_M(t) = \sum_{i=0}^{r-1} |FY^i| t^i$$

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Philosophy. If an enumerative polynomial is the Hilbert series of a Lefschetz algebra, then algebra explains its regularity.

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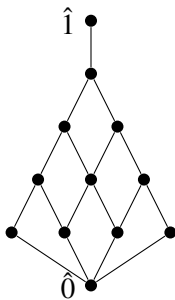
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Chow polynomials of posets

Bounded graded posets

Throughout, P is finite, bounded, and graded.



Feichtner–Yuzvinsky monomials

For a poset P a Feichtner–Yuzvinsky monomial is a monomial

$$m = x_{p_1}^{\ell_1} x_{p_2}^{\ell_2} \cdots x_{p_s}^{\ell_s} \in \mathbb{Q}[x_p : p \in P \setminus \{\hat{0}\}]$$

where

$$\hat{0} = p_0 < p_1 < \cdots < p_s \leq \hat{1} \quad \text{and} \quad 1 \leq \ell_i \leq \rho(p_i) - \rho(p_{i-1}) - 1.$$

Definition (Ferroni–Matherne–V.)

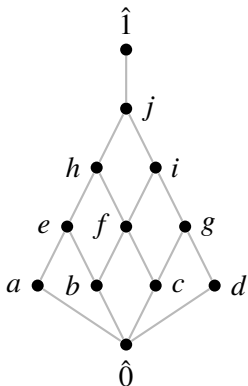
Let FY^k be the set of FY monomials of degree k of the poset P .
The Chow polynomial of P is

$$H_P(t) := \sum_{i \geq 0} |\text{FY}^i| t^i.$$

Building Feichtner–Yuzvinsky monomials

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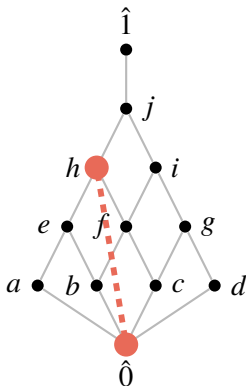
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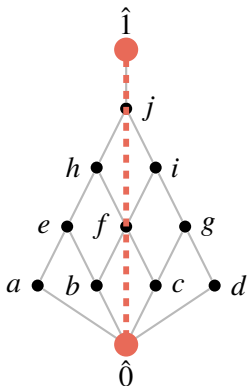


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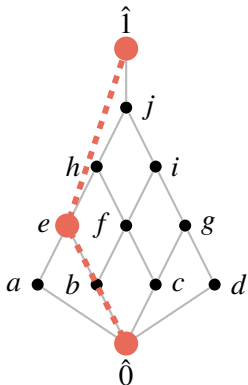


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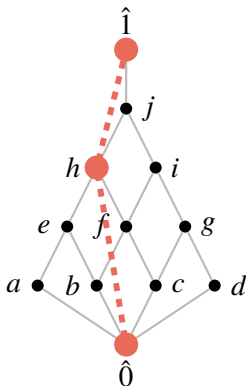
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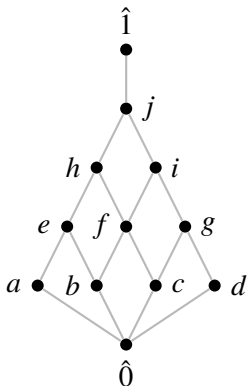
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Counting all FY monomials gives

$$H_P(t) = 1 + 7t + 12t^2 + 7t^3 + t^4.$$

Known properties

Theorem (Ferroni–Matherne–V.)

For every finite bounded weakly ranked poset P , $H_P(t)$ is monic, palindromic, and unimodal.

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These are exactly the first numerical properties one expects from a Hilbert series of a ring with Lefschetz properties.

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But we do **not** have a known canonical Lefschetz ring.

Main problem

Find a canonical algebra attached to P with Hilbert series $H_P(t)$ satisfying Poincaré duality and Hard Lefschetz.

Main results

Existence

Theorem (Schweitzer-V.)

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It proves that there is no obstruction for this ring to exist for arbitrary posets.

This does not construct the conjectural **canonical** Chow ring for P .

***O*-sequences and SI-sequences**

A sequence $g = (g_0, \dots, g_m)$ is an *O*-sequence if there is a monomial order ideal M such that

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Example:

$$M = \{1, x, y, x^2, xy\} \quad \Rightarrow \quad g = (1, 2, 2).$$

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Let $h = (1, h_1, \dots, h_d)$ be palindromic and unimodal. Define

$$g = (1, h_1 - 1, h_2 - h_1, \dots, h_{\lfloor d/2 \rfloor} - h_{\lfloor d/2 \rfloor - 1}).$$

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For us

$$h(P) = (|FY^0|, |FY^1|, \dots,)$$

encodes the coefficients of $H_P(t)$.

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$$g = (1, h_1 - 1, h_2 - h_1, \dots, h_{\lfloor d/2 \rfloor} - h_{\lfloor d/2 \rfloor - 1}).$$

For us

$$h(P) = (|FY^0|, |FY^1|, \dots,)$$

encodes the coefficients of $H_P(t)$.

Example:

$$h = (1, 7, 12, 7, 1) \rightsquigarrow g = (1, 6, 5)$$

O-sequences and SI-sequences

A sequence $g = (g_0, \dots, g_m)$ is an *O*-sequence if there is a monomial order ideal M such that

$$g_i = |\{m \in M : \deg m = i\}|.$$

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Definition

An SI-sequence is a palindromic unimodal sequence h such that g is an *O*-sequence.

This is enough: the g -theorem

Theorem (McMullen, Billera–Lee, Stanley)

For $h = (1, h_1, \dots, h_d)$, the following are equivalent:

- *h is an SI -sequence;*
- *h is the h -vector of a simplicial polytope;*
- *h is the Hilbert series of an algebra with the Kähler package generated in degree one.*

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Since $h(P)$ is already known to be palindromic and unimodal, it remains to show:

$g(P)$ is an O -sequence.

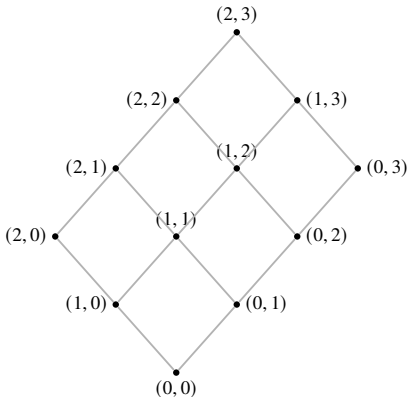
Strategy of the proof

Plan.

1. Order FY monomials by divisibility.
2. Build a combinatorial analogue of a Lefschetz decomposition.
3. Identify its primitive elements. They form a set SFY .
4. They are exactly $|FY^k| - |FY^{k-1}| = g_k$.
5. Prove that these form a monomial order ideal SFY .

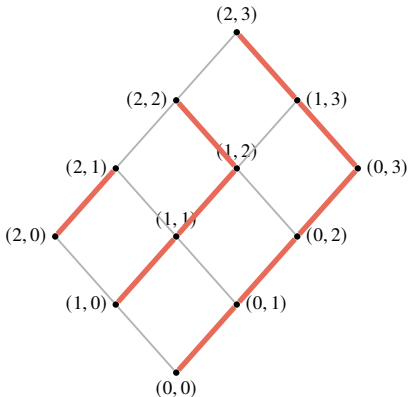
Symmetric chain decompositions

A SCD of a poset P is



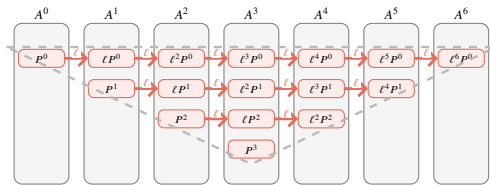
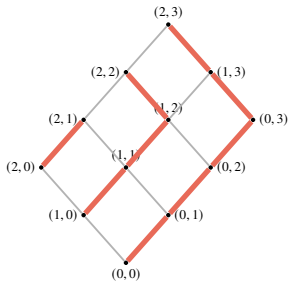
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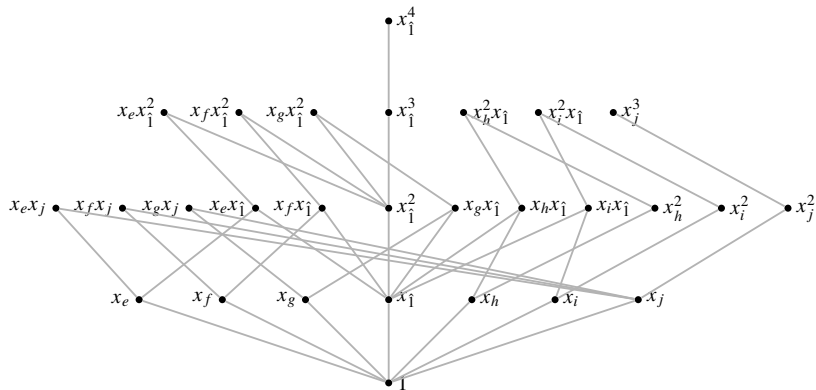


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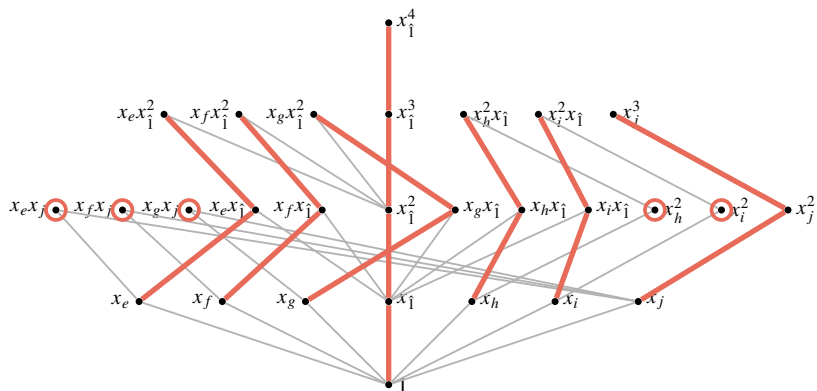


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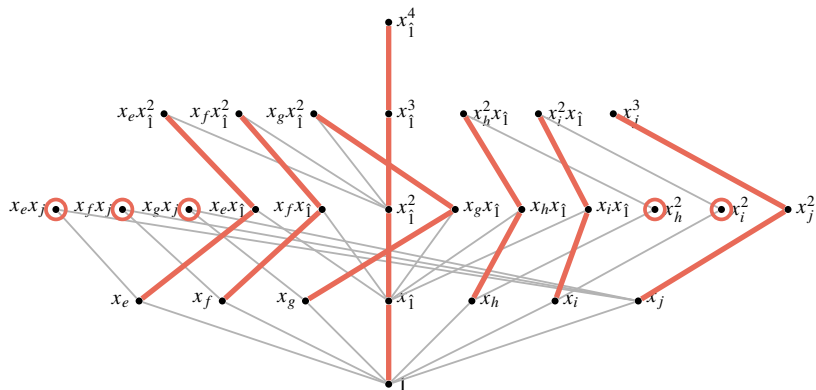


$$m' \leq m \Leftrightarrow m' | m.$$

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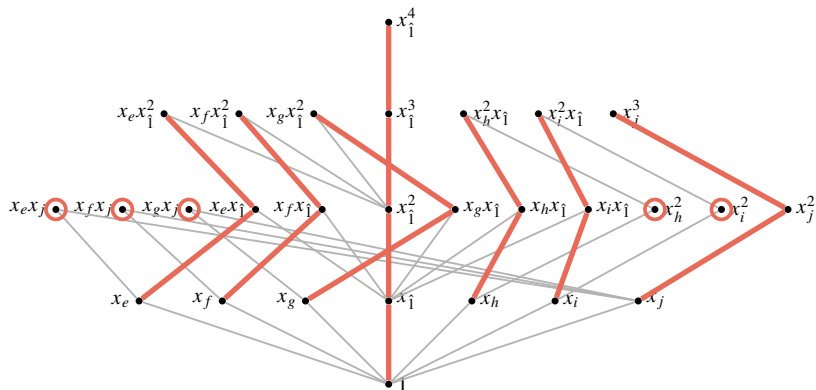


Example



$$\deg(\text{initial elements}) = (1, 7 - 1, 12 - 7)$$

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The special set SFY

This gives a distinguished set of initial monomials.

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A monomial

$$m = \prod_{k=1}^s x_{p_k}^{\ell_k} \quad \hat{0} = p_0 < p_1 < \cdots < p_s < \hat{1}$$

belongs to SFY if for each k ,

$$1 \leq \ell_k \leq \min \left(\rho(p_k) - \rho(p_{k-1}) - 1, \rho(P) - \rho(p_k) - 2 \sum_{i=k+1}^s \ell_i \right).$$

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For the running example,

$$\text{SFY} = \{1, x_e, x_f, x_g, x_h, x_i, x_j, x_e x_j, x_f x_j, x_g x_j, x_h^2, x_i^2\}.$$

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Proof idea.

If $m \in \text{SFY}$, any $m' | m$ still satisfies the defining inequalities.

Decreasing an exponent is straightforward. Removing a variable merges two rank jumps, so one must check that the new bound is compatible with the two old bounds.

Conclusion of the proof

The set SFY is such that

- It is a monomial order ideal.
- Its number of elements in degree k is

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- $h(P)$ SI-sequence $\implies H_P(t)$ is a Lefschetz Hilbert series.

Some consequences

Theorem (Schweitzer–V.)

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Theorem (Schweitzer–V.)

If P has rank at most 6 (sharp), then $H_P(t)$ is log-concave.

Palindromicity and unimodality always hold, but log-concavity does not hold in general.

What remains open

Main open direction

Find a canonical algebra $\mathrm{CH}(P)$ such that

$$\mathrm{Hilb}_{\mathrm{CH}(P)}(t) = H_P(t)$$

and $\mathrm{CH}(P)$ satisfies the full Kähler package.

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The present result says that the desired Hilbert series is always compatible with such an algebra. The g -theorem could even produce an explicit polytope!

But: this is not the same as a natural construction. The existence proof forgets the combinatorics of P . Consequently, it does not provide canonical classes, natural maps, or new poset invariants beyond the Hilbert series.

Thank you! 😊

A canonical decomposition

Theorem (Braden–Huh–Matherne–Proudfoot–Wang)

$$\underline{\text{CH}}(\text{M}) = \underline{\text{IH}}(\text{M}) \oplus \bigoplus_{\emptyset < F < E} \psi^F \left(\underline{\text{CH}}_{\text{M}|F} \otimes \underline{\text{J}}_{\text{M}/F} \right).$$

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Theorem (Ferroni–Matherne–V.)

$$\text{H}_P(t) = \frac{f_P^{\text{rev}}(t) - f_P(t)}{t - 1} + \sum_{\hat{0} < p < \hat{1}} \text{H}_{\hat{0}p}(t) \frac{f_{p\hat{1}}^{\text{rev}}(t) - t f_{p\hat{1}}(t)}{t - 1}$$

where f_{M} is the KL polynomial of M .

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Theorem (Ferroni–Matherne–V.)

$$H_P(t) = 1 + t \sum_{\rho(p) > 1} H_{\tau([\hat{0}, p])}(t).$$